

PICOFLOW · REQUIREMENTS

PicoFlow requirements

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DOCUMENT CONTROL

Document ID	PCF-REQ-001	Version	0.5
Status	Draft	Date	2026-10-01
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Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/picoflow		
Source	docs/03-requirements.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.5	2026-10-01	Amish Chadha	Status from PCF-CAL-001 v0.3 on the constructable design (PCF-DDR-003); R15 reported against the value-engineering target
0.4	2026-09-25	Amish Chadha	Recommendations accepted by Amish (DDR-002)
0.3	2026-09-25	Amish Chadha	Apply Amish's 2026-09-25 decisions (PCF-DDR-001), R15 to \$450, add R17 penstock surge, status from PCF-CAL-001
0.2	2026-09-25	Amish Chadha	First measurable requirements for TRL 2, with status against the concept estimates
0.1	2026-09-24	Amish Chadha	Initial scaffold

PicoFlow requirements

These requirements were set at TRL 2 and checked by calculation at TRL 3 in PCF-CAL-001 v0.3, on the constructable design of PCF-DDR-003. The status column gives the TRL 3 result; "not verifiable at TRL 3" means a bench or site test is needed. Decisions of 2026-09-25 are recorded in PCF-DDR-001 (the value-engineering target in R15 is USD 450, and R17 was added for the drainage-grade penstock) and PCF-DDR-002 (125 mm design-point penstock, R2 stated per head, R4 relaxed to 20 W, R14 definition confirmed, R15 margin accepted). Targets are still not validated with users or at a site.

The **design point** used throughout is 2.0 m of gross head (forebay water surface to nozzle centerline) and 10 L/s through two nozzles, with a 20 m run of 125 mm PVC penstock (decided by Amish, 2026-09-25, PCF-DDR-002).

ID	Requirement	Target	Verification (TRL 3 or later)	Status (PCF-CAL-001)
R1	Operating head range	Generates from 1.0 to 3.0 m gross head with the same runner, changing only nozzle inserts	Performance calculation across the range	Met; DC bus 19.7 V at 1.0 m
R2	Flow range	5 to 10 L/s at 1.0 m and 5 to 15 L/s from 2.0 m gross head, through interchangeable nozzle inserts, one or two jets	Nozzle sizing calculation	Met; 1.3 to 11.0 L/s at 1.0 m, 1.9 to 15.6 L/s at 2.0 m
R3	Output at the design point	80 W or more DC into the battery at 2.0 m and 10 L/s	Efficiency chain calculation; later bench test	Met, 82.5 W with the 34 mm inserts; 81.7 W at exactly 10 L/s (1.7 W margin)
R4	Output at the lowest head	20 W or more DC into the battery at 1.0 m and 7 L/s	Efficiency chain calculation	Met, 25.5 W
R5	Water-to-wire efficiency	40 % or more at the design point, from gross hydraulic power to DC into the battery	Calculation; later test	Met, 41.4 %
R6	Daily energy	1.8 kWh or more per day at the design point, running 24 h	Calculation	Met, 1.98 kWh
R7	Battery charging and load control	Charges a 12 V nominal battery; holds the runner within 10 % of its best speed while charging; diverts 100 % of output to the dump load within 1 s when the battery is full or disconnected	Controller design review; later bench test	Not verifiable at TRL 3; dump load 1.76 times the largest output
R8	Runaway and touch voltage	Every part survives indefinite runaway at 3.0 m head with no damage; no exposed conductor above 60 V DC in any state, including runaway	Speed and voltage calculation; design review	Met by calculation: hardware clamp keeps the bus at 48 V or less; 78.0 V only on a double fault, inside the enclosed 100 V DC side; generator overspeed rating to confirm
R9	Printable parts	Runner and nozzles print on a 220 x 220 x 250 mm build volume in PETG or nylon; runner prints in 24 h or less	Model check; slicer estimate	Met (estimate); runner 200 x 200 x 50 mm, about 21 h; nozzle 167 x 130 x 244 mm
R10	Debris tolerance	Intake screen aperture 6 mm or less; smallest nozzle 20 mm or more; screen self-cleans by overflow	Design review	Met by design
R11	Serviceability	Runner or nozzle inserts replaced with hand tools in 30 min or less; bearings in 60 min or less; bearings and generator above the spray zone	Design review; later timed trial	Not verifiable at TRL 3; bearings above the spray by layout
R12	Service life	Bearing L10 life 20,000 h or more; runner 1 year (8,760 h) or more before replacement	Bearing life calculation; material review	At risk ; bearings about 8.0×10^6 h, printed runner life unknown
R13	Setting height		Model check	Met, 280 mm in the model

ID	Requirement	Target	Verification (TRL 3 or later)	Status (PCF-CAL-001)
		Nozzle centerline 0.3 m or less above normal tailwater, so little head is lost below the jet		
R14	Portability	Heaviest single item 15 kg or less; turbine unit (items 1 to 6 and 11) 25 kg or less, carried to site by two people	Mass estimate	Met, 24.5 kg with its fixings; generator 7.0 kg (definition confirmed by Amish, PCF-DDR-002)
R15	Cost	Turbine kit (all items except penstock and battery) against a USD 450 value-engineering target, a hypothetical control target rather than a limit (Amish, 2026-10-01); salvaged-motor variant documented	Priced BOM (bom/ bom.csv)	Over the USD 450 value-engineering target by USD 84: USD 534 with a new generator; USD 464 with a salvaged motor (PCF-CAL-001 v0.3)
R16	Stream protection	Takes no more than half of the dry-season stream flow and returns all water to the same stream; intake passes fish-safe overflow	Site survey; design review	Not verifiable at TRL 3 (site-dependent)
R17	Penstock surge	Peak pressure at the valve (static plus surge) 50 kPa or less at 3.0 m head, with a valve that cannot close in under 10 s	Surge calculation	Met, 33.4 kPa with a multi-turn gate valve

■ Requirements not met or at risk

No requirement is not met by calculation on the constructable design (PCF-CAL-001 v0.3):

- **R2** is stated per head (5 to 10 L/s at 1.0 m, 5 to 15 L/s from 2.0 m), because 15 L/s at 1.0 m would need 61 mm jets on a 150 mm pitch circle.
- **R3 and R5** are met (81.7 W at exactly 10 L/s, 41.4 %), now with the 90 mm inlet valve's reducer and expander counted. The margin on R3 is 1.7 W, so it depends on the runner reaching the assumed 75 %. A full-bore valve would give 85.6 W; the valve bore is open decision N6 in the design decisions register (PCF-DEC-001).
- **R4** was relaxed from 30 W to 20 W; the calculation gives 25.5 W.
- **R12 (runner life) remains at risk.** Only a TRL 4 test can close it.
- **R15** is reported against the value-engineering target (Amish, 2026-10-01): the constructable kit is USD 534, USD 84 over the USD 450 target. Cost drivers and savings worth trying are in PCF-DEC-001.

■ Budget scope

The USD 450 value-engineering target covers the turbine kit: runner, shaft, bearings, coupling, generator, housing, manifold and nozzles, valve, forebay and screen, frame, equipment post, rectifier, controller, dump-load and clamp resistors, wiring, pipe stands and fixings. The penstock and the household battery are site- and household-specific and are costed separately (decided by Amish, 2026-09-25, PCF-DDR-001 D1).

■ Assumptions

The full list is in PCF-CAL-001 section 2. The main ones are:

- Penstock: 125 mm drainage PVC, SN8 (117.6 mm bore), 20 m, with a rounded entrance, screen, two bends and a gate valve; branches 90 mm.
- Nozzle velocity coefficient 0.97.
- Runner jet-to-shaft efficiency 75 % for a first printed design, below the 87 to 91 % reached in laboratory work on an optimized low-head Turgo ([Williamson et al., 2013](#)).
- Generator about 10 rpm per volt of rectified DC, 12 W fixed loss at 340 rpm, 0.8 Ω ; rectifier 1.6 V drop; buck converter 94 %.
- Drainage pipe joints good for 50 kPa, to be confirmed with the supplier.

Safety: R8 and R17 are safety requirements. They are met on paper only; neither the clamp nor the pipe rating has been tested.